

A Full Five-Cycle-Double-Cover Boundary Relation for a 288-Vertex Six-Pole

Atharva Vaidya

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Resolution status, novelty status, and trust boundary. The standard Five-Cycle Double Cover Conjecture remains open. This note proves a computer-assisted boundary lemma for one concrete six-pole and deduces, by a short human argument, a positive theorem for every trivalent network of copies of that pole. It is neither a proof nor a disproof of the general conjecture, and it does not concern the orientable variant. The novelty claim is provisional. Searches made before release did not locate this exact D_5 boundary relation, but the literature contains substantial adjacent work on FiveCDC superpositions and multipoles. Specialist prior-art review may show that the network theorem is known or subsumed by a stronger result. OpenAI Codex agents, under Atharva Vaidya’s direction, selected the experiment, wrote the programs, discovered and proved the network consequence, checked the artifacts, and drafted this note. The independent checker is implementation-independent, not independent of AI. No human peer review is claimed. A human author must check the proof and provenance, rerun or reimplement the finite audit, review prior art, and accept scholarly responsibility before journal or arXiv submission.

Abstract

Let

$$D_5 = \binom{\{0, 1, 2, 3, 4\}}{2}$$

be the ten duads, represented by their five-bit characteristic vectors. A D_5 -labelling of a cubic graph is a five-cycle double cover exactly when the xor of the three incident labels is zero at every vertex. We compute the complete six-port boundary relation of a concrete 288-vertex $(3, 3)$ -pole retained from the $g = 10$ construction of Karabáš, Máčajová, Nedela, and Škoviera.

The relation is maximal subject to the global parity obstruction: an ordered boundary word $(b_0, \dots, b_5) \in D_5^6$ extends through the pole if and only if $b_0 \oplus \dots \oplus b_5 = 0$. The positive direction is certified by 571 explicit internal labellings, one for every orbit of the 62,560 admissible boundary words under coordinate permutations. An independently written standard-library checker exhausts the one-million-word universe and directly checks every certificate; no negative SAT conclusion is used.

The full relation has a simple all-size consequence. In every finite network obtained by attaching all ports of any number of copies of the pole to trivalent junctions, the connector-junction incidence multigraph is cubic bipartite. Three perfect matchings give compatible triangle-duad labels at every junction and a zero-xor boundary word at every pole. Consequently every such network has a five-cycle double cover.

1 The exact claim and its scope

A *cycle* here means an Eulerian edge-subset: every vertex has even degree in it. It may be empty or disconnected. A *five-cycle double cover* is an indexed family $C_0, \dots, C_4$ of such subsets in which every edge occurs in exactly two members. Allowing empty members makes “five” mean “at most five.” Parallel edges, when present, are distinct edge objects. The pole used below is loopless; the network construction may create repeated macro-incidences but no incidence loops.

The standard conjecture, attributed to Celmins and Preissmann, says that every finite bridgeless graph has such a cover [2]. It remains open. In particular, the theorem in this note only eliminates a concrete superposition family from a counterexample search.

Karabáš et al. construct cyclically 5-edge-connected snarks of prescribed girth and oddness two by superposition [1, Theorem 5.1]. Our retained F_{10} pole is the concrete $g = 10$ instance built with the 70-vertex Harries (3, 10)-cage: the construction first forms their graph K_M , then deletes the two specified trivalent vertices u_1, u_3 , retaining their incident edges as two ordered three-port connectors. The resulting pole has 288 vertices, 429 internal edges, and six semiedges.

For reproducibility, this note treats the exact topology in `f10-atom.txt` as the formal definition. Its SHA-256 digest is

484503e6318f419457e5160e9f3f7605002621661e82456d57f200c8b6750742.

The first line records 288, 429, and 6; the second line gives the six port vertices, grouped as (0, 2, 3) and (13, 15, 16); the remaining lines are the internal edges. The package records the construction function from which this frozen topology was exported; the certificate theorem itself depends only on the frozen topology.

2 Duad labellings and FiveCDC

Represent each $a \in D_5$ by its characteristic vector in $\mathbb{F}_2^5$. Thus every a has Hamming weight two.

Lemma 2.1 (duad equivalence). *Let G be a finite loopless graph. The following data are equivalent:*

1. *five Eulerian edge-subsets $C_0, \dots, C_4$ covering every edge exactly twice;*
2. *a map $\lambda : E(G) \rightarrow D_5$ such that*

$$\bigoplus_{e \ni v} \lambda(e) = 0 \quad \text{for every } v \in V(G). \quad (2.1)$$

Proof. Given the five subsets, label an edge by the two indices of the subsets which contain it. The i -th coordinate of (2.1) says exactly that C_i has even degree at v . Conversely, from a duad labelling put $C_i = \{e : i \in \lambda(e)\}$. Weight two gives exact double coverage, and the coordinate equations give the Eulerian condition. $\square$

For a cubic vertex, (2.1) says that the three incident duads form a coordinate triangle. For example,

$$01 \oplus 02 \oplus 12 = 0. \quad (2.2)$$

For a pole, a boundary word records the labels assigned to its semiedges. It is *extendible* if labels can be assigned to all internal edges so that (2.1) holds at every pole vertex, counting its incident semiedge.

3 The complete boundary relation

Write P for the frozen 288-vertex pole and

$$R_P = \{(b_0, \dots, b_5) \in D_5^6 : (b_0, \dots, b_5) \text{ is extendible through } P\}.$$

Theorem 3.1 (full F_{10} boundary relation).

$$R_P = \{(b_0, \dots, b_5) \in D_5^6 : b_0 \oplus \dots \oplus b_5 = 0\}. \quad (3.1)$$

Proof. For necessity, xor (2.1) over all 288 pole vertices. Every internal edge appears twice and cancels, leaving precisely the six boundary labels.

The converse is a finite, positive certificate. There are 10^6 ordered words in D_5^6 , of which 62,560 have total xor zero. The symmetric group S_5 acts simultaneously on the five coordinates of all six duads. Direct canonicalization partitions the 62,560 admissible words into 571 orbits.

The frozen file `s5-representatives.json` contains one canonical word from each orbit. The corresponding row of `boundary-witnesses.jsonl` contains 435 duads: first the labels of the 429 internal edges in frozen order, then the six boundary labels. For each of the 571 rows, the checker verifies that every entry is a duad, that the claimed boundary is the orbit representative, and that the three labels incident with each of the 288 vertices xor to zero. A permutation of the five coordinates carries a verified witness to a witness for every other word in the same orbit. Hence every word satisfying (3.1) extends. $\square$

Remark 3.2 (what is and is not trusted). The proof uses no solver’s assertion of unsatisfiability. CaDiCaL was used only as a producer of positive labellings. The checker is an independently written Python program using only the standard library. It reconstructs the entire one-million-word universe, recomputes the 571 orbits, and directly checks 248,385 edge labels and 164,448 vertex equations. A full replay also recompiles and reruns the producer. The witness file has SHA-256 digest

3e7328b4d02a459f6d1f85b8f357aa5ee70c38d81138bcea5b8ecaef84b2961d.

Corollary 3.3 (emergent port symmetry). *The boundary relation R_P is invariant under the full symmetric group S_6 on the ports.*

Proof. Condition (3.1) is invariant under every permutation of its six entries. $\square$

This is symmetry of the relation, not a claim that every port permutation is induced by an automorphism of the underlying pole. Quotienting the relation by $S_5 \times S_6$ leaves 11 orbits.

4 Every trivalent network is positive

Definition 4.1. An F_{10} *trivalent network* is obtained from finitely many disjoint copies of P by attaching every port to exactly one new trivalent junction, with exactly three port edges incident with every junction. A junction may receive several ports from the same pole or from the same connector triple.

Theorem 4.2 (universal network theorem). *Every finite F_{10} trivalent network has a five-cycle double cover.*

Proof. Form a bipartite incidence multigraph H . Its left vertices are the two connector triples of every copy of P ; its right vertices are the trivalent junctions; and its edges are the individual port-to-junction incidences. Every left vertex has degree three because a connector has three ports, and every right vertex has degree three by construction. Repeated incidences give parallel edges, which are retained separately. Thus H is a cubic bipartite multigraph.

Every cubic bipartite multigraph is 3-edge-colourable. Indeed, for a set S of vertices in one part, its $3|S|$ incident edges end in $N(S)$, which can receive at most $3|N(S)|$ of them. Hall's condition holds, so H has a perfect matching. Removing it leaves a 2-regular bipartite multigraph, whose even cycles split into two further perfect matchings. Call the three colours a, b, c .

Map these colours to the triangle duads

$$q_a = 01, \quad q_b = 02, \quad q_c = 12.$$

At every junction each colour occurs once, so (2.2) gives the vertex condition. At every connector each colour also occurs once. The two connectors belonging to one pole therefore contain each of q_a, q_b, q_c exactly twice across their six ports, and their total xor is zero. By Theorem 3.1, this boundary word extends through that pole. Choose an extension independently for every pole. The resulting duad labelling satisfies (2.1) at every vertex of the expanded network. Lemma 2.1 converts it to a five-cycle double cover. $\square$

Remark 4.3. The positive labelling does not require the expanded graph to be simple, connected, bridgeless, or high-girth. Those restrictions matter only when the network is proposed as a candidate counterexample.

5 Reproducibility

The complete package is publicly available in the draft repository and is also located, relative to this source tree, in

`scratch/f10-boundary-relation-20260729/`.

The public review location is github.com/AtharvaVaidya/mathematics-research/pull/5. From that directory, the fast independent audit is

```
shasum -a 256 -c SHA256SUMS
python3 verify_boundary_relation.py \
    f10-atom.txt s5-representatives.json \
    boundary-witnesses.jsonl relation-summary.json
```

On the frozen package it reports

```
ordered_duad_words=1000000 xor_zero=62560 S5_orbits=571
positive=571 negative=0
directly checked 571 witnesses, 248385 edge labels,
and 164448 vertex xor equations
```

The package ledger itself has SHA-256 digest

5b3abc755f6879dc4c938d5da7f2ae507e47457cc735fe80c9ea58a4226ed6f6.

The command `python3 verify.py` performs an end-to-end replay. It first invokes the independent checker, then regenerates the orbit census, compiles the C++20 producer, reruns all 571 positive SAT queries, and checks the newly produced witnesses directly. This replay currently expects the CaDiCaL headers and static library in the Homebrew paths recorded by the script. The direct checker has no non-standard dependency.

6 Relationship to prior work and remaining questions

The pole topology comes from Karabáš et al. [1]. Their theorem concerns girth, oddness, colouring defect, and proper superposition; it does not state the exact D_5 -boundary relation proved here. More broadly, Liu, Hao, Luo, and Zhang give sufficient conditions for superpositions to have a five-cycle double cover and verify several snark families [2]. Máčajová and Škoviera develop the modern formalism of proper multipoles and superposition [3]. These are the closest sources found in the initial search and must be compared in depth before asserting novelty.

The contribution claimed by this draft is therefore deliberately narrow and provisional:

1. the explicit complete D_5 boundary relation of one frozen 288-vertex six-pole;
2. independently checkable positive witnesses for all boundary orbits;
3. the short cubic-bipartite incidence proof covering every trivalent network of copies of that pole.

The most natural next question is whether the same full boundary relation holds for a structural class of the F_g poles, rather than only this concrete $g = 10$ instance. A negative answer would also be useful: non-full boundary relations are potential building blocks for certificate-producing counterexample searches.

Acknowledgment and AI-use disclosure

This manuscript is an openly AI-assisted research artifact. OpenAI Codex agents made material contributions to the mathematical search, software, proof derivation, verification design, literature search, and prose. The repository history and frozen certificates are provided so that the claims can be assessed without trusting the AI system. The named human has not, by publication of this draft alone, represented that the work has passed independent expert review.

References

- [1] J. Karabáš, E. Máčajová, R. Nedela, and M. Škoviera, *Girth, oddness, and colouring defect of snarks*, Discrete Mathematics 345 (2022), Article 113040. doi:10.1016/j.disc.2022.113040.
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